

Date: _____

September 21, 2026 · 5:00 – 7:30 PM · Kai Santos

NOTES

MECH 191 — Metallurgy · Week 5 Notes · fill in blanks & key terms as you go

01 Stainless Steels

02 Stainless Families

03 Cast Irons

SLIDE 2 *Class Agenda*

Portfolio Handout — Module 4

NOTES

MECH 191 — Metallurgy · Week 5 Notes · fill in blanks & key terms as you go

MECH 191 — Course & Unit Roadmap

Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
Materials Fundamentals	Steel & Alloys	Testing & Quality	Manufacturing	Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	Wk 4 Sep 14 Carbon & Alloy Steels	Wk 7 Oct 5 Mechanical Testing	Wk 10 Oct 26 Casting	Wk 14 Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	Wk 5 ← here Sep 21 Stainless & Cast Irons	Wk 8 Oct 12 NDT Methods	Wk 11 Nov 2 Forming	Wk 15 Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	Wk 9 Oct 19 Quality Standards	Wk 12 Nov 9 Machining	Wk 16 Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 • FINAL EXAM

SLIDE 4 MECH 191 — Course & Unit Roadmap

NOTES

Topics

- What makes stainless steel corrosion-resistant — the passive chromium oxide layer
- The four stainless steel families and how microstructure defines each one
- Austenitic (300s): FCC, Ni-stabilized, non-magnetic, best corrosion resistance
- Ferritic & martensitic (400s): BCC/BCT, magnetic, hardenable grades
- Cast iron as a distinct iron-carbon family — not a low-quality steel
- Gray, white, ductile, and malleable cast iron — how carbon form drives properties
- When and why to choose cast iron over steel

Reference Material

Moniz 5th — Ch. 16 & 17
Cast Irons / Stainless Steels

Kalpakjian 6th — Ch. 6
Ferrous Metals and Alloys

Iron-Carbon Phase Diagram (Interactive)

AISI/SAE Steel Designation Codes

NOTES

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

- 1 What single threshold separates stainless steel from ordinary alloy steel — and why does that exact percentage matter?
- 2 A coastal facility needs a tank for chloride brine: 304 or 316? What is the deciding factor?
- 3 Two castings from the same iron-carbon melt: one becomes gray iron, one white iron. What determined which was which?
- 4 Why is gray cast iron standard for engine blocks and machine bases, despite having low tensile strength?
- 5 A martensitic stainless weld cracks in the heat-affected zone. What most likely went wrong, and how do you prevent it?

SLIDE 6 Core Questions

NOTES

All stainless steels share one feature: a minimum of 10.5% chromium by weight. That single threshold is what separates stainless from ordinary alloy steel.

- Chromium reacts with atmospheric oxygen to form a thin, stable Cr_2O_3 layer on the surface — invisible, only a few nanometers thick
- This passive layer acts as a barrier, preventing oxygen and moisture from reaching the steel beneath
- Self-repairing: if scratched or abraded, the layer reforms spontaneously in the presence of oxygen
- Destroy the passive layer (with chlorides, oxygen-poor environments, or mechanical damage in an aggressive medium) and corrosion can proceed rapidly
- 10.5% Cr is the threshold — below this, not enough Cr_2O_3 forms to maintain a complete, stable passive layer

≥ 10.5% Cr

Passive layer forms. Now classified as stainless steel.

Austenite stabilizer

Keeps FCC structure at room temp → non-magnetic, ductile, tough.

Pitting resistance

Improves resistance to chloride pitting (e.g., 316 vs. 304).

Hardenability

Enables hardening by heat treatment → martensitic grades.

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Same passive chromium oxide layer — very different microstructures. The balance of alloying elements determines which phase is stable at room temperature, and that phase determines everything.

Austenitic	Ferritic	Martensitic	Duplex
300 series	400 series	400 series	Various
FCC Non-magnetic · Cold work only Cr: 18–20% / Ni: 8–12%	BCC Magnetic · Not hardenable Cr: 10.5–30% / Ni: No Ni	BCC/BCT Magnetic · Quench & temper Cr: 10.5–18% / Ni: Low or none	Austenite + Ferrite Semi-magnetic · Not by HT Cr: 21–26% / Ni: 4–8%
Nickel stabilizes the FCC austenite phase all the way to room temperature. Most widely used family — best combination of corrosion resistance, formability, and weldability. Cannot be hardened by heat treatment.	High chromium content stabilizes the BCC ferrite structure. No nickel makes these grades less expensive. Moderate corrosion resistance, lower toughness in thick sections and at low temperatures.	Higher carbon content allows martensite to form on quenching. Can be hardened to high strength — but at the cost of lower corrosion resistance and toughness compared to austenitic grades. Requires preheat and PWHT to weld.	Mixed austenite/ferrite microstructure — higher strength than austenitic, better resistance to stress corrosion cracking (SCC). Increasingly common in oil & gas, chemical processing, and marine environments.
304 (general purpose) · 316 (+ Mo for chloride resistance)	430 (most common) · 439 · 446	410 · 420 · 440C	2205 · 2507
Food equipment, chemical vessels, marine hardware, kitchen appliances, medical devices	Automotive trim and exhausts, kitchen appliances, architectural panels, dishwasher liners	Cutlery, surgical instruments, valve stems, pump shafts, turbine blades	Subsea pipework, chemical plant, heat exchangers, desalination equipment

SLIDE 8 *Stainless Steel Families — Microstructure Is Everything*

NOTES

Austenitic Stainless — The Workhorse Family (300 Series)

Austenitic grades account for over 70% of all stainless steel produced. 304 and 316 are the two grades a technician will encounter most often.

304

General-Purpose Austenitic

18% Cr / 8% Ni — the most widely used stainless steel in the world

- Excellent corrosion resistance in most atmospheric, food, and chemical environments
- FCC structure — very ductile and tough, excellent formability
- Non-magnetic in annealed condition (can become slightly magnetic after cold work)
- Good weldability — low carbon 304L grade available for thicker weld sections to prevent sensitization
- Cannot be hardened by heat treatment — only by cold work
- Susceptible to chloride pitting and stress corrosion cracking (SCC) in chloride environments

Uses: sinks, countertops, food processing, chemical tanks, medical trays, architectural panels

316

Marine & Chemical Grade

18% Cr / 10% Ni / 2-3% Mo — the upgrade for chloride environments

- Molybdenum addition dramatically improves resistance to chloride pitting and crevice corrosion
- The grade of choice for marine environments, seawater, and halide-containing process streams
- Same FCC structure and non-magnetic behavior as 304
- 316L (low carbon) is preferred for welded assemblies to prevent sensitization
- Higher cost than 304 — specify it when 304 has failed in service or chlorides are known to be present
- Used in pharmaceutical and food processing where frequent washdowns with chlorine-based cleaners occur

Uses: marine fittings, chemical plant, pharmaceutical vessels, coastal architecture, lab equipment

SLIDE 9 Austenitic Stainless — The Workhorse Family (300 Series)

NOTES

Ferritic & Martensitic Stainless — The 400 Series

Both are magnetic BCC-based grades. Ferritic cannot be hardened; martensitic can — but both trade corrosion resistance for other performance priorities.

Ferritic (e.g. 430)

BCC structure • Magnetic • No nickel • NOT hardenable by heat treatment

- BCC ferrite phase at all temperatures — high chromium (10.5–30%) stabilizes BCC and prevents transformation to austenite
- No nickel = lower cost than austenitic grades
- Moderate corrosion resistance — good for mild environments, not suitable for chloride-rich or highly aggressive conditions
- Limited toughness, especially at low temperatures and in thick sections
- Good oxidation resistance at elevated temperatures — useful in exhaust systems
- Magnetic — can be detected and separated with magnets (useful in recycling/fabrication)

Martensitic (e.g. 410, 420)

BCC/BCT structure • Magnetic • Hardenable • Higher carbon than ferritic

- Higher carbon content allows martensite to form when austenite is quenched — the same mechanism as tool steels, applied to a stainless grade
- Can be hardened to high strength and hardness by quench and temper — the only stainless family with significant heat treatment response
- Lower corrosion resistance than austenitic or ferritic grades (less Cr, more C)
- Magnetic in all conditions
- Requires preheat (typically 200–300°C) and post-weld heat treatment when welding — failure to do so causes hard, brittle heat-affected zones prone to cracking
- 410 (general purpose) • 420 (higher C, cutlery) • 440C (maximum hardness)

Welding Note:

Ferritic: limited weldability in thick sections — grain growth reduces toughness. Martensitic: MUST preheat; PWHT strongly recommended. Never weld martensitic in the hardened condition.

SLIDE 10 *Ferritic & Martensitic Stainless — The 400 Series*

NOTES

Cast iron is NOT steel that didn't make it. It is a separate family of iron-carbon alloys with its own unique microstructures, properties, and engineering roles.

- Carbon content above 2.14% — typically 2.5–4% plus significant silicon (1–3%)
- At these carbon levels, the excess carbon cannot stay in solid solution — it must precipitate out during solidification
- The FORM the carbon takes when it precipitates determines everything about the cast iron type
- Silicon content affects how carbon precipitates: more Si → graphite forms; less Si → iron carbide (cementite) forms
- Cast iron is poured into complex shapes — generally not wrought, rolled, or drawn into structural sections
- Melting point lower than steel (~1150–1200°C vs. ~1500°C) — easier to pour into complex molds
- Generally more brittle than steel — most grades cannot absorb significant impact loading

Property	Steel	Cast Iron
Carbon	0.02–2.14%	> 2.14% (typ. 2.5–4%)
Structure	Wrought (rolled, forged, drawn)	Cast (poured into molds)
Ductility	Good to excellent	Poor (most grades)
Toughness	Good — absorbs impact	Poor — brittle fracture
Machinability	Moderate	Excellent (gray iron esp.)
Damping	Low	Excellent (gray iron)
Weldability	Fair to good	Poor — very difficult
Cost/shape	Higher for complex shapes	Cheap for complex shapes

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The same iron-carbon composition can produce dramatically different materials depending on cooling rate and silicon content — which determines the form carbon takes.

White Cast Iron

Carbon locked as IRON CARBIDE (Fe_3C / cementite) — rapid cooling, lower silicon

- Rapid cooling (or low Si content) prevents graphite from forming — carbon stays as cementite (Fe_3C)
- Cementite is extremely hard and brittle — Rockwell C 65-70+ in some compositions
- Very high wear resistance — resists abrasive wear exceptionally well
- Not machinable by conventional cutting tools — must be ground
- Very brittle — fractures with little or no warning, cannot absorb impact
- Fracture surface appears white or light-colored (cementite is pale)
- Starting material for malleable cast iron — white iron is annealed to convert cementite to carbon rosettes

Mill liners, crusher plates, grinding media, pump wear parts,
starting stock for malleable iron

SLIDE 12 *Gray & White Cast Iron — Two Extremes*

NOTES

Both achieve improved ductility by changing the shape of the carbon phase — eliminating the sharp flakes that make gray iron brittle.

Magnesium or cerium added to the melt just before casting — graphite forms as SPHEROIDS, not flakes

- Crankshafts, gears, heavy-duty pipe, automotive suspension, wind turbine housings, pump bodies

White cast iron annealed at high temperature — cementite converts to rounded TEMPER CARBON rosettes

- Pipe fittings, hand tools, brackets, agricultural equipment, automotive parts (legacy)

NOTES

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Cast iron is chosen when its specific characteristics are an advantage, not a limitation. Understanding which property each type excels at is the key to correct grade selection.

Gray Iron Graphite Flakes	White Iron Iron Carbide (Fe ₃ C)	Ductile Iron Graphite Spheroids	Malleable Iron Temper Carbon Rosettes
Strength: Low tensile high compressive Damping: Excellent Machin.: Excellent Toughness: Poor	Strength: Very high comp. no tension Damping: Poor Machin.: Not machinable Toughness: Very poor	Strength: High — up to 900 MPa Damping: Moderate Machin.: Good Toughness: Good	Strength: Moderate Damping: Moderate Machin.: Excellent Toughness: Moderate
Choose when: Best vibration damping of any common engineering metal. Choose for machine bases, engine blocks, brake rotors, and anything where vibration absorption matters.	Choose when: Maximum wear resistance is the priority and brittleness can be tolerated. Ideal for mill liners, crusher plates, and grinding equipment.	Choose when: Need good mechanical properties at lower cost than steel castings. Crankshafts, gears, heavy pipe, and structural castings where ductile iron replaces steel forgings.	Choose when: Thin-section castings requiring ductility and machinability where ductile iron is difficult to cast. Fittings, hand tools, and small hardware.
Avoid when: Any application with significant impact or tensile loading. Avoid where fracture toughness is needed.	Avoid when: Structural applications. Any impact loading. Cannot be welded or machined conventionally.	Avoid when: Extreme impact or shock loading — still less tough than steel. Not suitable where welding is required.	Avoid when: Thick sections — process economics make ductile iron more attractive. Being phased out for most new applications.

SLIDE 14 *Cast Iron — Choosing the Right Grade*

NOTES

Stainless & Cast Iron — Fabrication Considerations

Stainless steels and cast irons each present specific challenges in the shop. Understanding why these requirements exist is more valuable than just following a checklist.

Sensitization in Austenitic Stainless

When austenitic stainless is held at 425–815°C (during welding), chromium carbides precipitate at grain boundaries — depleting nearby zones of Cr below 10.5%. This 'sensitized' zone is no longer corrosion-resistant. Solution: use low-carbon grades (304L, 316L) or stabilized grades (321, 347 with Ti/Nb). Always specify L-grade for welded fabrication in corrosive service.

Martensitic Stainless — Preheat & PWHT

Martensitic stainless must be preheated (typically 200–300°C) before welding to slow cooling and prevent martensite formation in the heat-affected zone before hydrogen can escape. PWHT (temper at 650–750°C) restores toughness. Welding in the hardened condition without these steps causes cracking — the HAZ becomes a brittle fracture initiation site.

Cast Iron Welding — Why It's Difficult

High carbon content means high carbon equivalent — virtually all cast irons have CE > 1.0, requiring extreme preheat (400–600°C for gray iron) or cold welding with nickel-based filler. The primary risk is white iron formation in the HAZ as carbon migrates and cementite forms during rapid cooling. Pre-heat slow-cool and stress relief are mandatory for structural welds in gray and ductile iron.

Stainless Contamination Control

Stainless steel passive layers are destroyed by contact with carbon steel particles (rust transfer), chloride-containing cleaners, and abrasive contamination. In fabrication: use dedicated stainless tools, discs, and wire brushes. Never use carbon steel wire brushes on stainless. Pickle and passivate welds and heat-affected zones on austenitic stainless after fabrication to restore full corrosion resistance.

SLIDE 15 *Stainless & Cast Iron — Fabrication Considerations*

NOTES

Core Questions — Answers

1 What threshold makes stainless 'stainless,' and why that exact value?

≥ 10.5% Cr. Below this, not enough Cr_2O_3 forms to maintain a complete passive layer. Above this, the layer forms spontaneously and self-repairs when scratched.

304 or 316 for a chloride-brine tank — and why?

316. The 2-3% Mo addition dramatically improves resistance to chloride pitting and crevice corrosion — exactly the failure mode seawater causes in 304.

Same melt, two castings: one gray iron, one white iron. What determined which?

Cooling rate and silicon content. Slow cooling + higher Si → graphite flakes form (gray iron). Rapid cooling + lower Si → carbon stays as cementite/Fe₃C (white iron).

4 Why gray cast iron for engine blocks despite low tensile strength?

Graphite flakes provide outstanding vibration damping and excellent compressive strength — the loads are compressive. Graphite also self-lubricates and aids machining.

5 Martensitic stainless HAZ cracks after welding. What went wrong?

Missing preheat (200–300°C) and/or PWHT. Without preheat, hydrogen is trapped in brittle martensite. Without PWHT (temper 650–750°C), the HAZ remains hard and fractures.

SLIDE 16 Core Questions — Answers

NOTES

Key Takeaways

1 Stainless steel is a family — the grade determines the microstructure

Specifying 'stainless' without a grade is insufficient. The microstructure (FCC austenite, BCC ferrite, BCT martensite) determines magnetic behavior, hardness response, toughness, and corrosion resistance in each service environment.

10.5% chromium creates the passive layer — everything else is fine-tuning

Ni stabilizes austenite. Mo improves chloride resistance. C enables hardening. Si and Cr content determine which phase is stable. Every grade decision traces back to what the passive layer needs to survive in the intended environment.

3 Cast iron's properties are determined by the form its excess carbon takes

Graphite flakes (gray) → damping and machinability but brittle. Spheroids (ductile) → strength and toughness. Carbides (white) → extreme hardness but very brittle. Temper carbon (malleable) → improved ductility after annealing.

Cast iron is not failed steel — it is a deliberate engineering choice

When damping, castability, or wear resistance is the priority, cast iron outperforms steel. The engineer chooses the material based on service requirements — a technician who understands why helps avoid misapplication.

5 Safety is not optional!

PPE is mandatory in the shop — no exceptions. We work with heat, sharp edges, and hazardous materials every lab session. Protect yourself and your classmates.

SLIDE 17 Key Takeaways

NOTES

Week 5 Resources

- ## Visualizers

- ## Coming Up — Week 6
- # Nonferrous Metals & Phase Diagrams

- Aluminum, copper, and titanium — why each family exists alongside steel
- Reading the iron-carbon phase diagram — phases, regions, and key transformation points
- From equilibrium to real-world cooling — how CCT diagrams show what actually forms
- Unit 2 Quiz covers Weeks 4-6 — review steel grades, heat treatment, and nonferrous metals

SLIDE 18 *Resources & What's Next*

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